

**Topics:** 15.1 Double Integrals over Rectangles; 15.2 Double Integrals over General Regions; 15.3 Double Integrals in Polar Coordinates

Please put away all electronic devices, including cell phones and calculators.

*This content is protected and may not be shared, uploaded, or distributed.*

1. Evaluate each double integral.

(a)  $\iint_R \left(y + \frac{x}{y^2}\right) dA$ , where  $R = \{(x, y) \mid 0 \leq x \leq 2, 1 \leq y \leq 2\}$ .

$$\begin{aligned} \iint_R (y + xy^{-2}) dA &= \int_1^2 \int_0^2 (y + xy^{-2}) dx dy = \int_1^2 \left[ xy + \frac{x^2 y^{-2}}{2} \right]_0^2 dy \\ &= \int_1^2 (2y + 2y^{-2}) dy = [y^2 - 2y^{-1}]_1^2 = 4 \end{aligned}$$

(b)  $\iint_R x \cos^2(y) dA$ , where  $R = [-2, 3] \times [0, \frac{\pi}{2}]$ .

$$\begin{aligned} \iint_R x \cos^2(y) dA &= \int_{-2}^3 \int_0^{\pi/2} x \cos^2(y) dy dx = \int_{-2}^3 \int_0^{\pi/2} x \left( \frac{1 + \cos(2y)}{2} \right) dy dx \\ &= \int_{-2}^3 \left[ \frac{x}{2} \left( y + \frac{1}{2} \sin(2y) \right) \right]_0^{\pi/2} dx = \int_{-2}^3 \frac{x}{2} \left( \frac{\pi}{2} \right) dx = \int_{-2}^3 \frac{\pi}{4} x dx = \left[ \frac{\pi}{8} x^2 \right]_{-2}^3 = \frac{9\pi}{8} - \frac{4\pi}{8} = \frac{5\pi}{8} \end{aligned}$$

(c)  $\iint_R \frac{1}{(2x + 3y)^2} dA$ , where  $R$  is the rectangle with vertices  $(0, 1)$ ,  $(1, 1)$ ,  $(1, 2)$ ,  $(0, 2)$ .

$$\begin{aligned} \iint_R \frac{1}{(2x + 3y)^2} dA &= \int_1^2 \int_0^1 (2x + 3y)^{-2} dx dy = \int_1^2 \left[ -\frac{1}{2} (2x + 3y)^{-1} \right]_0^1 dy \\ &= \int_1^2 -\frac{1}{2} (2 + 3y)^{-1} + \frac{1}{2} (3y)^{-1} dy = -\frac{1}{2} \int_1^2 \frac{1}{2 + 3y} - \frac{1}{3y} dy = -\frac{1}{2} \left[ \frac{1}{3} \ln |2 + 3y| - \frac{1}{3} \ln |y| \right]_1^2 \\ &= -\frac{1}{2} \left[ \left( \frac{1}{3} \ln |8| - \frac{1}{3} \ln |2| \right) - \left( \frac{1}{3} \ln |5| - \frac{1}{3} \ln |1| \right) \right] = -\frac{1}{6} (\ln(8) - \ln(2) - \ln(5)) \end{aligned}$$

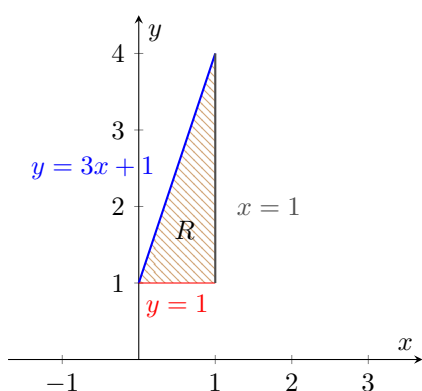
2. Find the volume of the solid that lies under the hyperbolic paraboloid

$$z = 3y^2 - x^2 + 2$$

and above the rectangle  $R = [-1, 1] \times [1, 2]$ .

$$\begin{aligned}
 V &= \int_{-1}^1 \int_1^2 (3y^2 - x^2 + 2) dy dx = \int_{-1}^1 [y^3 - yx^2 + 2y]_1^2 dx \\
 &= \int_{-1}^1 (9 - x^2) dx = \left[ 9x - \frac{x^3}{3} \right]_{-1}^1 = \frac{52}{3}
 \end{aligned}$$

3. Given the region  $R$  bounded by the lines  $y = 3x + 1$ ,  $x = 1$ , and  $y = 1$ , draw the region  $R$  and set up the double integral  $\iint_R f(x, y) dA$  as a Type I integral and as a Type II integral.



**Type I**

$$R = \{(x, y) \mid 0 \leq x \leq 1, 1 \leq y \leq 3x + 1\}$$

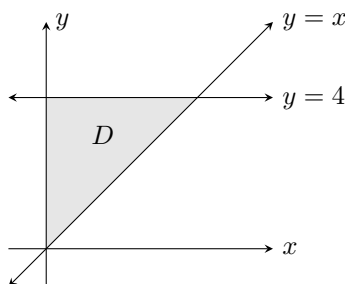
$$\int_0^1 \int_1^{3x+1} f(x, y) dy dx$$

**Type II**

$$R = \left\{ (x, y) \mid 1 \leq y \leq 4, \frac{y-1}{3} \leq x \leq 1 \right\}$$

$$\int_1^4 \int_{\frac{y-1}{3}}^1 f(x, y) dx dy$$

4. Evaluate  $\iint_D y^2 e^{xy} dA$ , where  $D$  is the region bounded by  $y = x$ ,  $y = 4$ , and  $x = 0$ .



Taking  $D$  as a Type I region, we get  $D = \{(x, y) \mid 0 \leq x \leq 4, x \leq y \leq 4\}$  and

$$\iint_D y^2 e^{xy} dA = \int_0^4 \int_x^4 y^2 e^{xy} dy dx.$$

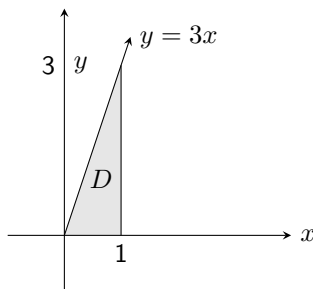
Taking  $D$  as a Type II region, we get  $D = \{(x, y) \mid 0 \leq y \leq 4, 0 \leq x \leq y\}$  and

$$\iint_D y^2 e^{xy} dA = \int_0^4 \int_0^y y^2 e^{xy} dx dy.$$

Note that integrating with respect to  $y$  first requires integration by parts, while integrating with respect to  $x$  first is more straightforward. So, if we evaluate the integral using  $D$  as a Type II plane region, after integrating with respect to  $x$ , we need to integrate the function  $ye^{y^2}$ , which can be done with the  $u$ -substitution  $u = y^2$ . Thus, we get

$$\int_0^4 \int_0^y y^2 e^{xy} dx dy = \int_0^4 (ye^{y^2} - y) dy = \frac{1}{2}[e^{16} - 17].$$

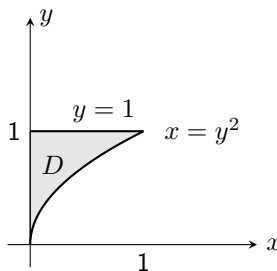
5. Find the average value of the function  $f(x, y) = xy$  over the triangular region with vertices  $(0, 0)$ ,  $(1, 0)$ , and  $(1, 3)$ .



The area of a triangle with base  $b$  and height  $h$  is  $A = \frac{1}{2}bh$ . So the area of the region is  $\frac{1}{2}(1)(3) = \frac{3}{2}$ . Then

$$f_{\text{avg}} = \frac{2}{3} \int_0^1 \int_0^{3x} xy dy dx = \int_0^1 3x^3 dx = \frac{3}{4}$$

6. (a) Sketch the region of integration for  $\int_0^1 \int_0^{y^2} f(x, y) dx dy$ .



- (b) Reverse the order of integration for the integral in part (a).

$$\int_0^1 \int_{\sqrt{x}}^1 f(x, y) dy dx$$

(c) Evaluate the integral from part (b) with the function  $f(x, y) = \sqrt{x} + 3y$ .

$$\begin{aligned} \int_0^1 \int_{\sqrt{x}}^1 (\sqrt{x} + 3y) dy dx &= \int_0^1 \left[ y\sqrt{x} + \frac{3}{2}y^2 \right]_{y=\sqrt{x}}^{y=1} dx \\ &= \int_0^1 \left( \sqrt{x} + \frac{3}{2} - x - \frac{3}{2}x \right) dx \\ &= \int_0^1 \left( \sqrt{x} + \frac{3}{2} - \frac{5}{2}x \right) dx \\ &= \left[ \frac{2}{3}x^{3/2} + \frac{3}{2}x - \frac{5}{4}x^2 \right]_0^1 \\ &= \frac{2}{3} + \frac{3}{2} - \frac{5}{4} = \frac{11}{12} \end{aligned}$$

7. Using polar coordinates, find the volume of the solid that lies under the cone

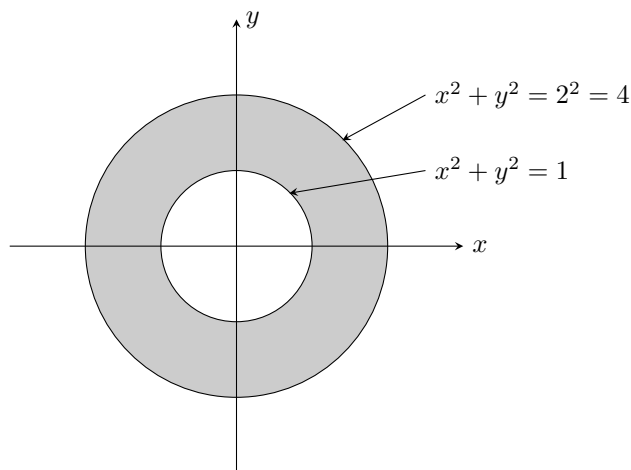
$$z = \sqrt{x^2 + y^2}$$

and above the annulus  $1 \leq x^2 + y^2 \leq 4$ .

The function  $z = f(x, y) = \sqrt{x^2 + y^2} \geq 0$ , so

$$V = \iint_D f(x, y) dA = \iint_D \sqrt{x^2 + y^2} dA$$

where  $D = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4\}$ .

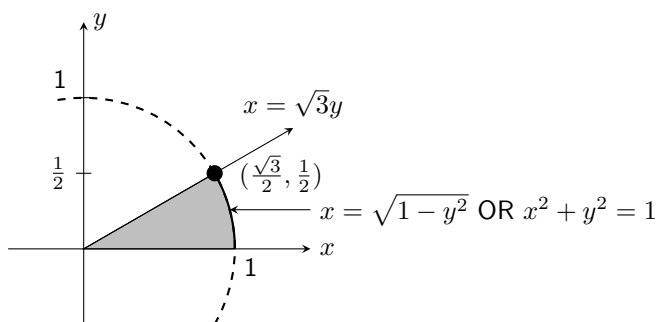


Converting to polar coordinates, we have  $D = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}$ . Recalling that  $r^2 = x^2 + y^2$  and  $dA = r dr d\theta$ ,

$$V = \int_0^{2\pi} \int_1^2 \sqrt{r^2} r dr d\theta = \int_0^{2\pi} \int_1^2 r^2 dr d\theta = \int_0^{2\pi} \frac{7}{3} d\theta = 2\pi \frac{7}{3} = \frac{14\pi}{3}$$

8. Evaluate  $\int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2 dx dy$  by changing to polar coordinates.

The region is given as a Type II plane region  $D = \{(x, y) \mid 0 \leq y \leq 1/2, \sqrt{3}y \leq x \leq \sqrt{1-y^2}\}$



In polar coordinates,  $D = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq \pi/6\}$ ,  $x = r \cos(\theta)$  and  $y = r \sin(\theta)$ , so the line  $x = \sqrt{3}y$  in polar coordinates is  $r \cos(\theta) = \sqrt{3}r \sin(\theta)$ . That is  $\tan(\theta) = \frac{1}{\sqrt{3}}$  and  $\theta = \frac{\pi}{6}$ . Then,

$$\begin{aligned} \int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2 dx dy &= \int_0^{\pi/6} \int_0^1 (r \cos(\theta))(r \sin(\theta))^2 r dr d\theta \\ &= \int_0^{\pi/6} \int_0^1 r^4 \cos(\theta) \sin^2(\theta) dr d\theta \\ &= \frac{1}{5} \int_0^{\pi/6} \cos(\theta) \sin^2(\theta) d\theta \\ &= \frac{1}{120} \end{aligned}$$

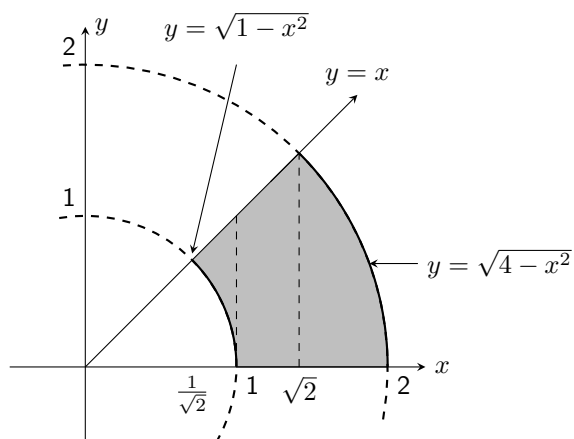
where the last integral is solved by the  $u$ -substitution  $u = \sin(\theta)$ .

9. Use polar coordinates to combine the sum

$$\int_{1/\sqrt{2}}^1 \int_{\sqrt{1-x^2}}^x xy dy dx + \int_1^{\sqrt{2}} \int_0^x xy dy dx + \int_{\sqrt{2}}^2 \int_0^{\sqrt{4-x^2}} xy dy dx$$

into one double integral, and evaluate the integral.

**Answer:**



Note that the domain of each integral in the problem statement above represents one of the vertical bands of the domain shaded in gray. Each region is written as a Type I plane region. We can simplify the problem by representing the region  $D$  in polar coordinates as  $D = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi/4\}$ . Then, the integral in polar coordinates is

$$\begin{aligned} \iint_D xy \, dA &= \int_0^{\pi/4} \int_1^2 r \cos(\theta) r \sin(\theta) r \, dr \, d\theta \\ &= \int_0^{\pi/4} \int_1^2 r^3 \cos(\theta) \sin(\theta) \, dr \, d\theta \\ &= \frac{15}{4} \int_0^{\pi/4} \cos(\theta) \sin(\theta) \, d\theta \\ &= \frac{15}{16} \end{aligned}$$

where the final integral is evaluated using the substitution  $u = \sin(\theta)$ .

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

### The Average Value of a Function of Two Variables.

The average value of a function  $f$  over a region  $R$  of area  $A(R)$  is

$$f_{\text{avg}} = \frac{1}{A(R)} \iint_R f(x, y) dA,$$

provided that  $A(R) > 0$ .

### Integrals over General Regions.

Suppose that  $f$  is continuous on a region  $D$ .

If  $D$  is a Type I region of the form

$$D = \{(x, y) \mid a \leq x \leq b, \ g_1(x) \leq y \leq g_2(x)\},$$

where  $g_1$  and  $g_2$  are continuous on  $[a, b]$  and  $g_1(x) \leq g_2(x)$  for all  $x \in [a, b]$ , then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

If  $D$  is a Type II region of the form

$$D = \{(x, y) \mid c \leq y \leq d, \ h_1(y) \leq x \leq h_2(y)\},$$

where  $h_1$  and  $h_2$  are continuous on  $[c, d]$  and  $h_1(y) \leq h_2(y)$  for all  $y \in [c, d]$ , then

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

**Double Integrals in Polar Coordinates.** Suppose that  $f$  is continuous on a region  $D$  in the plane, and that  $D$  is described in polar coordinates by

$$\alpha \leq \theta \leq \beta, \quad h_1(\theta) \leq r \leq h_2(\theta),$$

where  $h_1$  and  $h_2$  are continuous on  $[\alpha, \beta]$ ,  $0 \leq h_1(\theta) \leq h_2(\theta)$  for all  $\theta \in [\alpha, \beta]$ , and  $0 \leq \beta - \alpha \leq 2\pi$ . Then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta.$$

---

### Suggested Textbook Problems

Section 15.1: 1-49, 53-56

Section 15.2: 1-50, 55-79

Section 15.3: 1-49